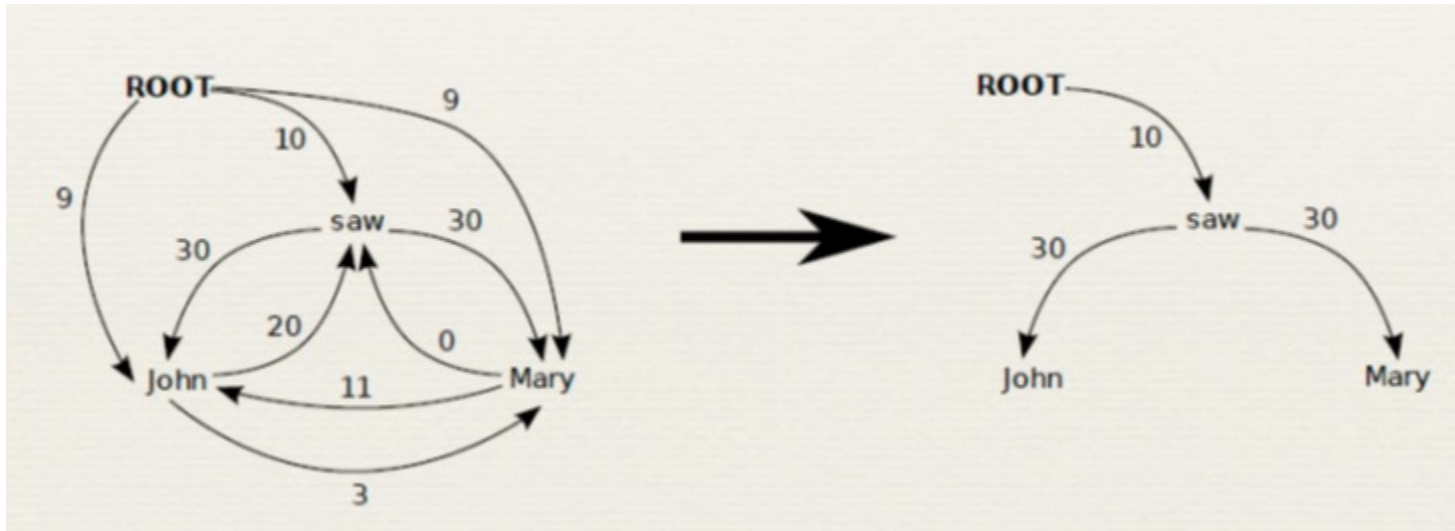


Graph-based Parsing

- Graph-based parsing addresses it as a structured prediction problem
- MST Parser:
 1. Score the arcs independently, based on how likely they are to appear in a parse
 2. Find the maximum directed spanning tree over the resulting weighted graph

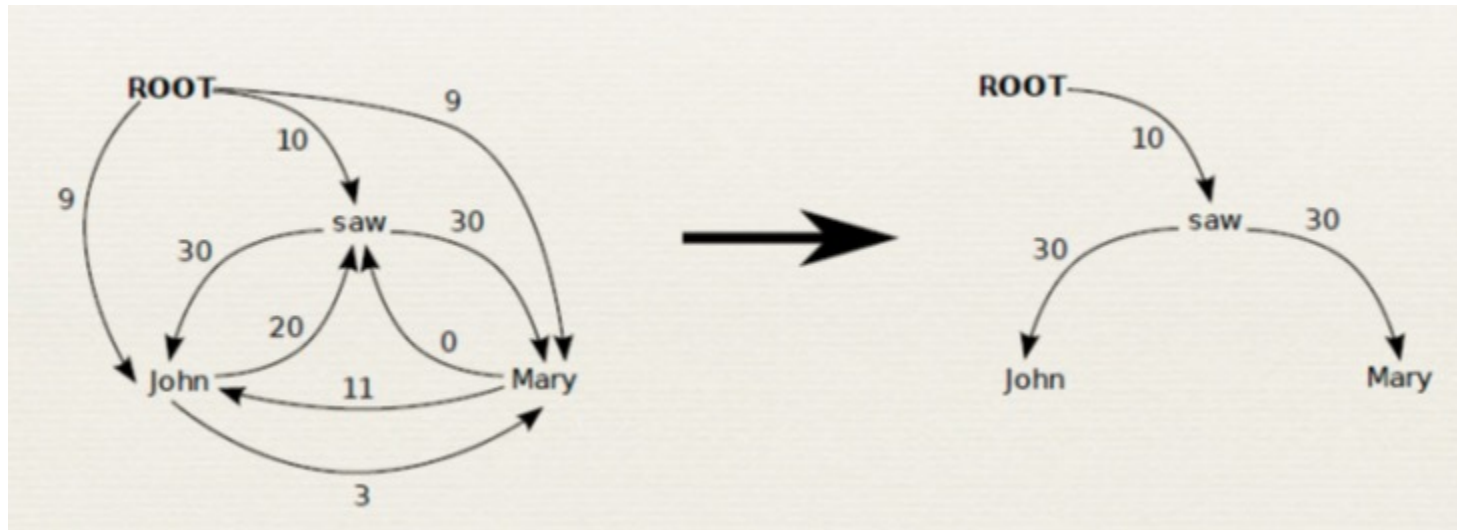


Online Large-Margin Training of
Dependency Parsers

R. McDonald, K. Crammer, and
F. Pereira, *ACL 2005*

Graph-based Parsing

- It is also possible to have a multi-graph with several possible edges between each two nodes
 - Useful when using labeled dependency parsing – an edge is needed for each possible labels
 - During inference, it is possible to prune away all edges but the maximal, reducing the multi-graph to a graph



Online Large-Margin Training of
Dependency Parsers

R. McDonald, K. Crammer, and
F. Pereira, *ACL 2005*

Features Used in Graph-based Dep Parsing



Features from McDonald et al.

- Identities of the words w_i and w_j and the label l_k

head=saw & dependent=with

Features Used in Graph-based Dep Parsing

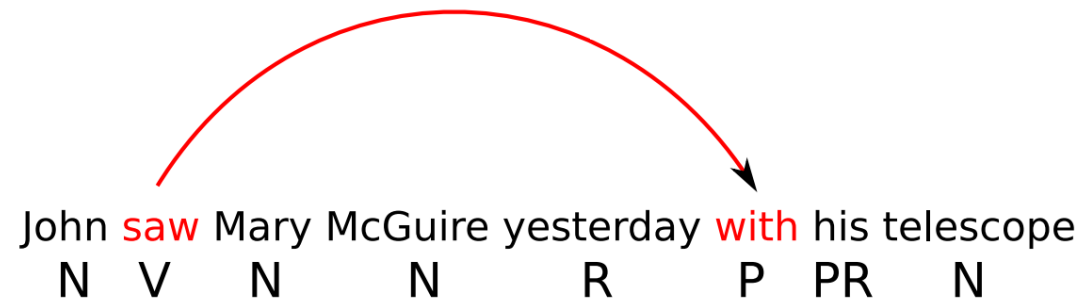


Features from McDonald et al.

- Part-of-speech tags of the words w_i and w_j and the label l_k

head-pos=Verb & dependent-pos=Preposition

Features Used in Graph-based Dep Parsing



Features from McDonald et al.

- Part-of-speech of words surrounding and between w_i and w_j

inbetween-pos=Noun
inbetween-pos=Adverb
dependent-pos-right=Pronoun
head-pos-left=Noun

...

Again conjoined with the label
(omitted from now on for brevity)

Features Used in Graph-based Dep Parsing



Features from McDonald et al.

- Number of words between w_i and w_j , and their orientation

arc-distance=3

arc-direction=right

Features Used in Graph-based Dep Parsing



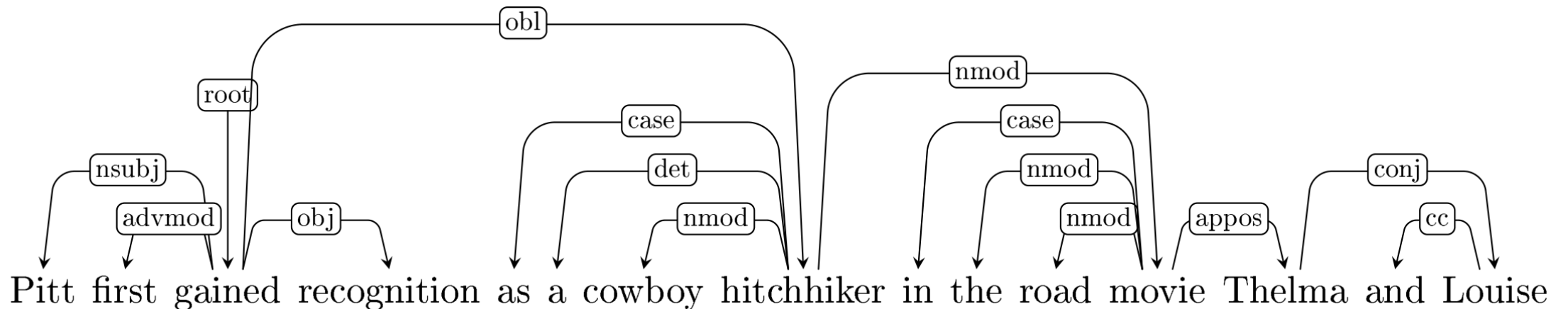
Label features

arc-label= L

And combinations of all these features...

Graph Based Parsing – Higher Order Features

- MST inference works well when the model factorizes over edges
- However, it has been proven useful to add features that involve several edges, which renders inference intractable
 - These are called “higher order features”



Higher-order Parsing Models

- Higher-order parsing models don't assume that the score of a tree factorizes over edges, but over sets of edges
- For instance, the “grandchild model” assumes that

$$\text{score}(T) = \sum_{(i,j,k) \in V^3 \mid (i,j) \in T \& (j,k) \in T} \text{score}(i,j,k) + \sum_{(i,j) \in T} \text{score}(i,j)$$

$$\text{score}(i,j,k) = \theta_1^t \cdot \phi_1(i,j,k, x_{1:n})$$

$$\text{score}(i,j) = \theta_2^t \cdot \phi_2(i,j, x_{1:n})$$

- We add the grandchild features to the standard MST ones terms (second summation)

Higher-order Parsing Models

- The sibling model with labels would look like this:

$$score(T) = \sum_{(i,j,k) \in V^3 | (i,j) \in T \& (i,k) \in T} score(i,j,k) + \sum_{(i,j) \in T} score(i,j)$$

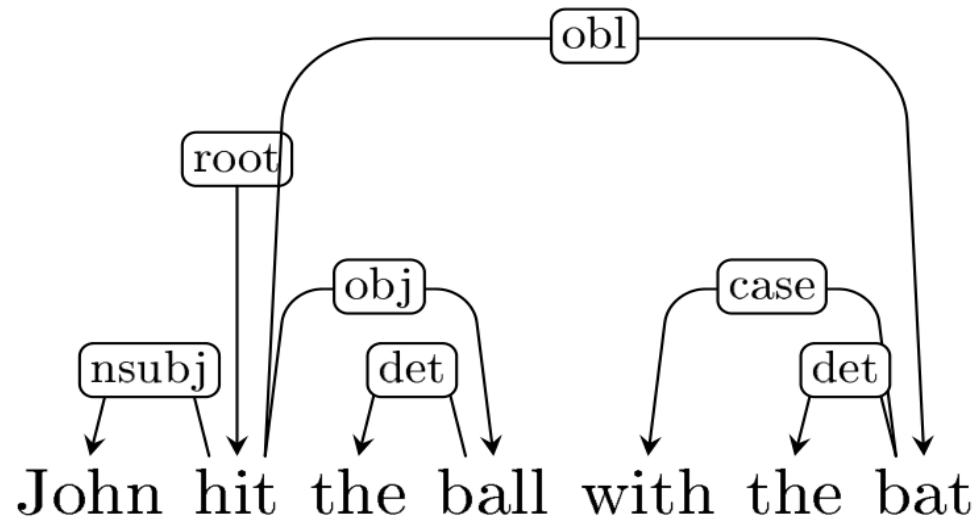
$$score(i,j,k) = \theta_1^t \cdot \phi_1(i,j,k, x_{1:n})$$

$$score(i,j) = \theta_2^t \cdot \phi_2(i,j, x_{1:n})$$

- Inference (finding the highest scoring tree given θ) is NP-complete. However, there are approximate algorithms.

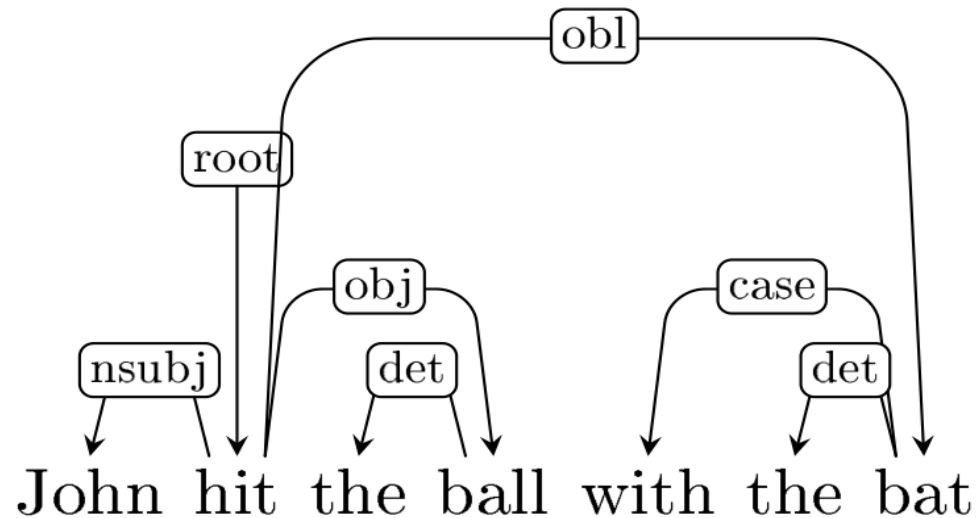
Graph Based Parsing – Higher Order Features

- **Sibling features** are features defined over two edges that have the same parent
- They can be helpful, because once you know “hit → ball” is an edge, “hit → bat” becomes more likely (over “ball → bat”)



Graph Based Parsing – Higher Order Features

- **Grandparent features** are features defined over two edges that form a chain
- They can be helpful, because “hit → bat” becomes much more likely once “bat → with” is included



Valence Features

- Verbs tend to appear in specific patterns, in terms of their number of arguments, their order and their prepositions
 - Subject gave Object₁ Object₂
 - Subject gave Object₂ to Object₁
 - Object₁ was given to Object₂
 - It was Object₂ that was given to Object₁
 - ...
- Encoding what sub-categorization frame is associated with each verb can promote subcategorization frames that appeared during training
- This feature is defined over all children of the verb and some of their children (higher than second-order)

Graph-based Parsing – Higher Order Features

- A lot of interesting machine learning research has been done on building learning models for higher-order features
- A few examples:
 - Xavier Carreras, 2007. Experiments with a Higher-Order Projective Dependency Parser. In CoNLL
 - Yuan Zhang, Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2014. Greed is good if randomized: New inference for dependency parsing. In EMNLP
 - Ilan Tchernowitz, Liron Yedidsion and Roi Reichart. 2016. Effective Greedy Inference for Graph-based Non-Projective Dependency Parsing. In EMNLP